Algorithms Syllabus

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Introduction: The Role of Algorithms in Computing and Real-Life Applications.

Complexity Analysis: Complexity Analysis of Algorithms, Worst Case, Best Case and Average Case.

Sorting Algorithms: Divide and Conquer Approach, Merge Sort and Quick Sort Algorithm, Complexity Analysis, Worst and Average Case Analysis, Heap Construction Algorithm, Heap Sort, Application of Heap: Priority Queue, Decision Tree Model and (Worst Case) Lower Bound on Sorting, Sorting in Linear Time - Radix Sort, Bucket Sort,Counting Sort, etc.

Graph Algorithms: Representation of Graphs, Breadth First Search, Depth First Search,Minimum Spanning Tree, Kruskal and Prims Algorithm.

Shortest Path: Dijkstra's Algorithm, Bellman-Ford Algorithm. Floyd Warshall Algorithm.

Searching Algorithms: Binary Search Trees, Balanced Binary Search Trees, A VL Trees and Red-Black Trees, Hashing. Priority Queues, Heaps.

Dynamic Programming: Longest Common Subsequence (LCS), Matrix Chain Multiplication (MCM), Knapsack Problem.

Greedy Algorithm: Greedy Algorithm, Activity Selection Problem, Huffman Codes and its application, Knapsack problem, Tree Vertex Splitting.

Recurrences & Backtracking: Recurrences, NP-Hard and NP-Complete Problems,Backtracking, N-Queen Problem.

Branch and Bounds: Basic Idea and Control Structure of Branch and Bound, FIFO Branch and Bound, LC Branch and Bound.

Reducibility between Problems and NP-completeness: Lower Bound Theory, Discussion of Different NP-Complete Problems like Satisfiability, Clique, Vertex Cover,

Independent Set, Hamiltonian Cycle, Travelling Salesman Problem (TSP),
Knapsack, Set
Cover, Bin Packing.

Computational Geometry: Computational Geometry, Line Segment Properties, Convex Hull, Graham Scan Algorithm of Convex Hull.